Toph Jones

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EDUCATION

Ohio University | Athens, OH — Anticipated Graduation: May 2029
Bachelor of Science in Electrical Engineering — GPA: 3.5

EXPERIENCE

Ohio University | Athens, OH — August 2025 – Present
SAE Baja Project Lead

- Designed and led development of an Active Rear Steer system using a Teensy 4.0 microcontroller, VNH5019 motor driver, and real-time PID control loop. Reduced turning radius by 20% and improved lap time by 8% as proven by onboard sensors
- Led CAN Test Bus sub team, designed a custom PCB in Altium Designer to validate CAN transmissions across the vehicle, accelerating integration of three subsequent board designs Made and presented Progress reports to show how my team was affecting the Baja team as a whole and how effective the team is.
- Delivered written and oral progress reports communicating hardware status and performance metrics to multi-disciplinary stakeholders

Taco Bell | Medina, OH — September 2024 – August 2025
Shift Lead

- Led shifts of up to 12 team members during peak hours, coordinated, drive-through and front of house to maintain fast service.

LEADERSHIP AND INVOLVEMENT EXPERIENCE

The Institute of Electrical and Electronics Engineers | Athens, OH — August 2025 – Present
Secretary

- Provided administrative support to first year students in Ohio University's IEEE Chapter, promoting interest in the program and maintaining positive student relations during a remote academic year
- Recorded and distributed meeting attendance, maintaining organized records that help other officers make decisions about workshops and budget.

COURSEWORK & PROJECTS

Rear Steer Buggy Integration | SAE Baja — Spring 2026

- Developed C++ firmware on Teensy 4.0 implementing cascading PID control for real-time steering angle correction at up to 1 kHz update rate
- Created A custom schematic and printed circuit board using Altium Designer using techniques such as differential pairs for the CAN lines.
- Integrated CAN bus communication to receive wheel speed sensor data and relay actuator commands across the vehicle network

- **CAN Test Bus** | SAE Baja — Fall 2026
- Designed a 2-layer board in Altium Designer featuring a CAN transceiver, LM7805 voltage regulation, and test headers for live bus monitoring
- Used Teensy 4.0 to absorb information from CAN Bus and display onto a Screen using I^2C.

HONORS AND RECOGNITION

Live Mas Scholarship | Live Mas Foundation — May 2025
Altium Essential Certification | Altium Designer — December 2025

SKILLS

Technical: C++, Python, Altium Designer, Java